

Traffic Risk and Congestion Intelligence System for Unstructured Indian Traffic



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By

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ABBREVIATION LIST

Abbreviation	Expansion
BoT-SORT	Robust Associations Multi-Pedestrian Tracking (tracker)
CMC	Camera Motion Compensation
COCO	Common Objects in Context (dataset)
CV	Cross-Validation
FGVD	Fine-Grained Vehicle Detection (dataset)
IDD	India Driving Dataset
IoU	Intersection over Union
mAP50	Mean Average Precision at an IoU threshold of 0.50
MOT	Multi-Object Tracking
OOB	Out-Of-Bag (error estimate)
px	Pixels
RF	Random Forest
SSM	Surrogate Safety Measure
TTC	Time To Collision
XGBoost	Extreme Gradient Boosting
YOLO	You Only Look Once (detector family)

ABSTRACT

In the Indian roads traffic is diverse mostly consisting of two and three wheeled vehicles with no proper following of lane discipline rules. These reasons make it difficult to deploy or use the traffic analytics techniques that are mainly designed for orderly traffic where all traffic rules are followed. We aim to utilise computer vision, machine learning and object tracking techniques to develop a pipeline that estimates the congestion levels and risk levels from a traffic video.

We used a YOLO11m to detect different classes of vehicles on Indian roads by fine tuning it using Fine-Grained Detection dataset. We achieved an average precision of 89.2%. we utilised object tracking BoT-SORT tracker to track vehicles in videos.

Traditional Threshold methods are not largely present and cannot be applied for every environment we used k means and silhouette score to decide the threshold. We calculated Silhouette score for 2,3,4,5 clusters and cluster 2 was selected with silhouette score of 0.825. We then divided a video frame into five features displacement ratio,speed variance,proximity interaction, deceleration events and the calculated congestion score.

After reviewing multiples papers and reviews there are no proper risk labels available we used a three way clustering method that generates safe moderate and high risk labels> Random Forest and XGBoost were used to train and automate the task with as split made at video level so that no clip appears in more than one partition and achieved a test accuracy of 0.979 we calculated F1 score of 0.985, 0.966 and 0.954.Both the models took displacement ratio as highest priority. The main Contribution is we have developed a pipeline where threshold is derived than being assumed.

Keywords: unstructured traffic, object detection, multi-object tracking, congestion estimation, surrogate safety measures, unsupervised labelling.

PREFACE

Road safety and driving is one of the most serious problem India is facing to help overcome it we need to understand the traffic patterns and traffic flow and detect the hotspots with the developments in domains like computer vision object tracking and machine learning techniques we can analyse these trends and patterns that could not have been possible, by manual human observations. With this Project we have tried to score Congestion flows and risk scores by analysing a normal traffic video from Indian streets.

Computer vision Tools and technologies like YOLO was used and it was fined tuned on FGVD dataset so that it can accurately identify different Indian vehicle classes tracking mechanisms like BoT-SORT was to track and extract vehicle movements and were divided into five features for every vehicle across short windows. And machine learning techniques to identify underlying patterns and learn to automate the process with this project we have tried to create and automate the process of identifying the risk and congestion patterns that can help authorities and planners to design and implement Measures to reduce the traffic risk and help towards a safer condition on Indian roads

1. Introduction

Road transport in India imposes a burden of death, injury and delay that is measured in hundreds of thousands of lives each year. The Ministry of Road Transport and Highways recorded 4,87,707 road accidents in 2024, in which 1,77,175 people were killed and 4,71,441 injured, an average of 485 deaths every day (Ministry of Road Transport and Highways, 2026). Those crashes are also becoming more lethal rather than less: severity, measured as deaths per hundred accidents, rose from 21.6 in 2005 to 36.3 in 2024. The distribution of the harm is what makes the Indian setting distinctive. Two-wheeler occupants account for 46.2 per cent of the fatalities and pedestrians for a further 20.6 per cent, so roughly two-thirds of those killed were not inside a car at all. Congestion imposes a parallel cost in the same cities: peak-hour congestion in Delhi, Mumbai, Bengaluru and Kolkata has been estimated at 149 per cent against an Asian average of 67 per cent, at a cost of about 22 billion US dollars a year (Boston Consulting Group, 2018), and the Economic Survey 2025-26 puts the annual loss to congestion across India's largest metropolitan cities at 1.47 lakh crore rupees (Government of India, 2026). Reducing either burden requires knowing where and why conflicts occur, which is a measurement problem before it is an engineering one.

For the most of the history the measurement came from crash records Crashes are unavoidable they happen randomly and they are generated only after a it happened .A junction can be dangerous for years before it has before it accumulates enough recorded collisions to become statistically visible and by then there might be occurrence of several accidents This has pushed traffic-safety research towards surrogate safety measures, which are quantities observable in ordinary traffic that stand in for risk without waiting for a crash. Time to collision and post-encroachment time are the established members of this family, and the approach continues to be refined

Almost every method existing carries an assumption that is rarely stated because, in the settings where these methods were developed, it is simply true. The assumption is that traffic is structured: that vehicles occupy discrete lanes, that they are broadly similar in size and dynamics, that overtaking is an event with a beginning and an end, and that a vehicle's position can meaningfully be described relative to a lane centreline

India consists of very diverse traffic flows. Vehicles of different classes like cars autorickshaw, buses, bicycles etc are on the same road stretch with usually little or almost no lane discipline the movement of vehicles is continuous and cannot be divided or organised into lanes.

Such Kind of movements that usually does not follow lane discipline is unstructured traffic and traditional methods do not work where they assume several standards are followed. Lane occupancy cannot be calculated using standard methods where the lane discipline is not consistently followed, we in this study have tried to create and identify solution that helps us build analytical models by combining different computer vision, tracking and machine learning models.

1.1 Aim and objectives

The aim of this study is to determine whether ordinary unstructured video of Indian traffic can be used to predict the congestion and risk assessments using thresholds that are derived from data rather than fixed in advance.

Objectives

- Develop object detector on Indian vehicle classes and multi object tracking framework.
- Identification of congestion threshold and estimation of congestion score.
- Estimation of live risk score based on selected features.
- Development of dashboard for visualization of congestion risk score.

2. Literature Review

Zand et al (2025) and team used ByteTrack and found that even weak detection that are removed can still hold enough information of a vehicle. Aharon et al (2022) and team in the same year published BoT-SORT it is multi object tracking algorithm it combines camera motion compensation appearance information and utilizes a *Kalman filter*. The selection of detectors is fine but for it to perform optimally training it on right conditions is important in 2019 Varma and team used the India Driving Dataset and found that European driving models don't perform well in Indian roads Varma and team achieved an accuracy of .09 on Indian vehicles. The FGVD dataset that was published by Khoba and team in 2022 arranges vehicles in different classes making identification of vehicles on Indian road condition easier.

Traditional threshold techniques and number cannot be used for Indian roads cannot be used directly as traditional methods assume certain rules are followed so we need a method

MacQueen.J (1987) introduced the k means clustering that assigns each one of the observations to its cluster that is closest to its mean

Since Kmeans itself cannot choose the number of clusters in Rousseeuw et al (1987) came up with Silhouette coefficient which weighs an observation closeness to its own cluster against the nearest alternatives.

Onelcin and Alver built a weaving-section score for occasional weaving at fixed merge points, which shows by contrast that Indian traffic needs weaving treated as a constant condition rather than an event. Khan et al (2024) observed that fatigue has drawn far more research attention than sudden lateral movement, naming precisely the gap this project sets out to fill.

Scoring risk is harder and the established methods were designed for other roads Abdel-Aty et al (2023) analysed the usual indicators like time-to-time collision and found that these measures handle angled movement poorly that predominantly exists in Indian roads.

On sudden braking and movements Dong (2023) published their report they noticed an abrupt shift in position or speed, covering hard braking and rapid lane changes, which is what the displacement-ratio and deceleration-event descriptors register. Xu, Liu, Yao and Lyu (2023) added that a lane change cannot be judged alone, because how near it brings a vehicle to others also matters, and that is why a proximity count accompanies the trajectory descriptors here.

Jinn, Lin and Lu (2025) built SSAnet which can detect and track vehicles and passes the result to classifier which produces low medium and high labels that were tested across various road scenes. It outperformed several existing models present it requires a large labelled dataset to train that is not present for india traffic.

kumar and Rubal (2021) studied and surveyed more than 100 studies on deep learnin for detecting conjestopn and its prediction they separated video-based speed estimation from image-based density estimation their conclusion was that both congestion and density together give us fuller picture rather that they either of these are used separately almost all the work they covered was on fixed cameras on structured roads and none in their studies addressed lane free traffic. but the finding itself establishes density and speed as the two signals a congestion measure should be built around, and that is the basis used here.

Lin (2025) used a combination of a vision language model with motion analysis to classify congestion on a five-point scale and found that combination of these performed better than

single signal methods conforming that multi-level congestion; labels are a meaningful direction Vision-language models carry a computational cost that does not suit the hardware available here, so the same multi-level outcome is reached through a lightweight partition instead.

Tong and Liang (2025) combined an improved YOLOv11 with an enhanced DeepSORT tracker on expressway footage, feeding flow, density and speed into a GRU with attention to warn of congestion before it develops, reaching over 95 per cent detection accuracy and 93.8 per cent tracking accuracy with vehicles partly occluding one another. The goal, forecasting congestion on lane-structured expressways, differs from classifying current congestion on Indian roads. The relevant finding is the tracking accuracy under heavy occlusion, since occlusion is constant in the footage used here.

Li and Chen (2025) predicted accident risk by combining a CNN extracting speed and acceleration from each trajectory, an LSTM tracking how risk evolves, and a graph neural network modelling how nearby vehicles affect one another, and found the combination beat models built from any single component. It was tested only on structured highways where normal movement is easy to define. Its central idea, that a vehicle's own motion and its interaction with others are complementary rather than alternative signals, is the principle behind pairing speed-variance and deceleration descriptors with a proximity descriptor here.

By reading all the literature review is What is missing is a system that takes detection and tracking as given, builds a congestion classifier and a risk score from quantities a bare video stream provides, infers every cut-off from the footage itself, and is then tested on real unstructured Indian traffic. That is the space this project occupies.

3. Study Area and Data

3.1 Study Area

The study area is India. Indian traffic differs from the structured traffic that is assumed by most of the literature. The vehicle classes are dominated by two and three wheelers. Lane

marking is frequently absent so any measure defined relative to a lane is unavailable. And lateral movement is continuous rather than occasional, so measures that treat weaving as a discrete event do not apply.

The project consists of 124 training videos recorded across Indian roads covering both fixed and moving camera viewpoints, resolutions

3.2 Training data: the FGVD dataset

For fine tuning the detector the Fine-Grained Vehicle Detection dataset was used which was published by center for visual information technology,IIIT Hyderabad (Khoba et al,2022), it consists of 5502 images captured from camera on moving car and has 24450 vehicle bounding boxes across 210 fine grained labels.This dataset was designed specifically for identifying different classes of vehicles on india roads.

3.3 Evaluation

The dataset that we took for the evaluation and training part consists of 124 videos.These videos are drawn from publicly released Indian road scene video collection and a set of long form videos captured from vehicle mounted cameras.

Total videos:124

Frame Rates present: 30.0 frames per second (116 videos), 25.0 fps (6), 59.94 fps (1), 29.97 fps (1)

Environments: Market footages, junctions, urban centers.

4. Methodology

4.1 Overall design

The methodology consisted of seven major stages:

- (i) dataset preparation,
- (ii) detector fine-tuning,
- (iii) vehicle detection and tracking,
- (iv) congestion estimation,
- (v) risk feature engineering,
- (vi) pseudo-labelling and supervised validation, and
- (vii) visualisation and dashboard development.

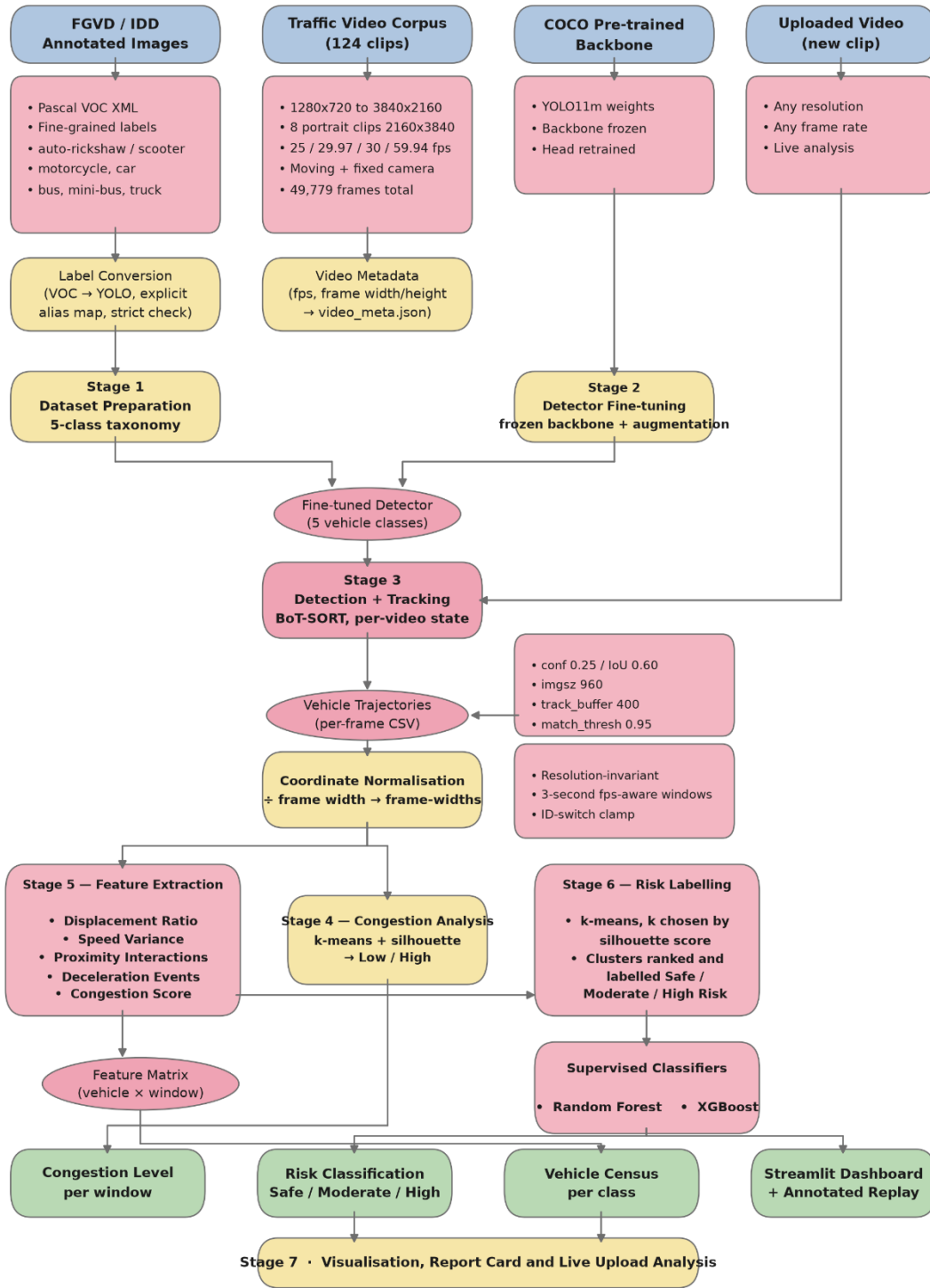


Figure 4.1 Seven-stage architecture of the pipeline

4.2 Stage 1: dataset preparation

Detection was done using yolov11m but it does not detect Indian vehicle class like autorickshaw etc so it was fine-tuned on FGVD but FGVD annotations carry labels in the form of vehicle type, manufacturer and vehicle model. But for our proposed pipeline that

manufacturer names and model's labels are not necessary for road level analysis. We converted the FGVD script into YOLO text format.

The first version of the code did not recognise most of the classes from our dataset and were dropped without an error. As it compared each label against the class names in FGVD exactly, so a label for example motorcycle_yamaha_R15 did not match motorcycle class.

The corrected version checks a label if it begins with a known class name. after the fix 24,450 annotations were retained across training, validation and testing.

Index	Class	Relevance
0	Car	four-wheeled category
1	Motorcycle	Dominant category by count; high lateral mobility
2	Auto-rickshaw	Three Wheeled category
3	Bus	Large footprint; strong effect on local density
4	Truck	Large footprint; slow acceleration

Table 4.1 Vehicle classes retained after collapsing the label hierarchy

4.3 Stage 2: detector fine-tuning

The choice of detector that was chosen was YOLO11m from the YOLO computer vision family as it is the medium variant of the family, which balances detection accuracy against inference cost. These models were fine-tuned on the FGVD dataset to suit the Indian environment. The input resolution was 960 by 960 pixels and data loader was set to zero so as to avoid a multiprocessing conflict.

4.4 Stage 3: detection and tracking

All the 124 videos from our dataset were run frame by frame using the fine-tuned detector and multi object tracking tool BoT-SORT was utilised for maintain and tracking the identity of vehicles across frames. The frames were processed as a stream rather than they be held in memory, so that the video length doesn't affect processing.

One csv per video was created with one row per tracked vehicle frame, frame index, track identifier, class, bounding-box centers, width and height and detection confidence. These files are the input to every later stage

Bot-SORT was used the fine-tuned detector it uses sparse optical flow to estimate the camera motion it needs enough matchable background points between consecutive frames. On footage from a two-wheeler the inter-frame displacement is large and image suffers distortion due to which estimate fails and tracker falls back to assuming a stationary camera.

4.5 Stage 4: congestion estimation

Each video was divided into windows of three seconds because the research paper contains several frame rates the window that we used was defined in seconds and converted to frames for each video by using its measured rate, like giving 75 frames for videos at 25fps, 90 frames for videos at 30 fps and 180 frames for videos at 60 fps. Three seconds window was selected as it is long enough for a manoeuvre to develop and it is short enough for conditions be treated as steady if longer seconds are selected then there will be many changes and we accurately cannot extract our information

Instead of fixing a threshold K-means clustering was used to calculate congestion values for every window the clustering value was selected by comparing distribution for cluster counts from two to five and each cluster from two to five was evaluated using the silhouette coefficient.

The best scoring cluster was selected and kept and the midpoint between two clusters centers is taken as threshold.

4.6 Stage 5: risk feature engineering

For each vehicle that is observed in each window five features were computed the table 4.2 gives the definitions and reasoning behind each.

Descriptor	Definition	Interpretation and antecedent
Displacement ratio	Straight-line distance between first and last position, divided by the total length of the path travelled	Close to one for straight, purposeful movement; falls as the path becomes indirect. Captures the continuous weaving Onelcin and Alver (2023) identify
Speed variance	Variance of the frame-to-frame speed of the bounding-box centre within the window	Low for steady progress, high where the vehicle alternates between accelerating and braking, following Dong et al. (2023)
Proximity interactions	Count of frames in which the vehicle centre lies within a derived distance of another tracked vehicle centre	Calibration-free analogue of the gap measures of Yang and Hong (2026); supports Xu et al. (2023) on nearby-vehicle context
Deceleration events	Count of consecutive-frame transitions in which speed falls by more than a derived threshold	Image-space counterpart of the deceleration-rate surrogate; hard braking is an established indicator (Boateng et al., 2023)
Congestion score	The window-level congestion value from Stage 4	Traffic-density context; Kumar and Raubal (2021) show density and speed together describe traffic state best

Table 4.2 The five risk descriptors

4.6.1 Deriving the proximity and deceleration thresholds

Out of the five features two features proximity and deceleration need a threshold to decide what counts as close and what is considered as sharp braking.

For proximity, the nearest-neighbour distance was computed for every tracked vehicle in every frame and k-means with two clusters fitted to the resulting distribution. The two centres fall in the bulk of the distribution and in its upper tail, and the midpoint between them becomes the threshold; the distribution is single-peaked rather than clearly bimodal, so the value is best read as a reproducible cut-off derived from the data rather than a boundary between two separate regimes; this is the Stage 4 procedure applied to a different quantity, validated the same way. For deceleration, clustering is less appropriate because speed drops form a single skewed distribution rather than two regimes, so the threshold was set at 1.5 standard deviations above the mean of all observed frame-to-frame speed reductions. This flags the upper tail of braking events without requiring a cluster structure the data does not have. Both values are reported in Chapter 5 and written to the parameters file.

4.7 Stage 6: pseudo-labelling and supervised validation

4.7.1 The labelling problem

Supervised risk classification needs risk labels, and none exist for this corpus; producing them by expert annotation would have meant reviewing thousands of windows. The alternative adopted is to generate labels by clustering. The five descriptors were standardised and k-means applied, with the cluster count chosen by silhouette validation, and the resulting clusters ordered by their mean descriptor profile and given the ordinal labels Safe, Moderate Risk and High Risk. The logic is that a cluster with low displacement ratio, high speed variance and frequent proximity and deceleration events describes the movement pattern the surrogate safety literature associates with elevated conflict risk. What this gives has to be stated precisely: a partition of the feature space into three regions with a defensible ordinal interpretation, not ground truth. That distinction governs how the results in Chapter 5 should be read.

4.7.2 Classifiers and regularisation

Two classifiers were trained to reproduce the partition: a Random Forest (Breiman, 2001) and a gradient-boosted tree ensemble (Chen & Guestrin, 2016). Two model families were used deliberately, since agreement between methods with different inductive biases is stronger evidence that the partition is learnable than one model's accuracy would be. This mirrors the hybrid design Uddin et al. (2025) recommend, where deep learning handles perception and simpler interpretable methods handle the final decision. Both were regularised conservatively, because a few thousand rows with five features is small enough for an unconstrained tree ensemble to memorise.

The forest was set at depth 8 with a minimum of five samples per leaf and ten per split, drawing the square root of the feature count at each split, and validated with its out-of-bag score. XGBoost used a learning rate of 0.05 with early stopping after twenty rounds without improvement, subsampled 0.8 of rows per boosting round, and applied L1 and L2 penalties of 0.1 and 1.5 against the held-out validation partition. Both were additionally checked with five-fold stratified cross-validation.

4.8 Stage 7: visualisation and dashboard

For the results two visual outputs were produced. The first is charts and graphs figures that cover results like feature importance, confusion matrix, risk class distribution and cluster validation, plus clips figures that cover the risk and congestion timeline. The second is an interactive dynamic dashboard that was built using streamlit it has a command center with all the information about number of vehicles in each class it has a model page that contains all the graphics and charts like feature importance and confusion matrices it also contains pages like data explorer and an video analysis option that accepts uploading a new video and runs the full pipeline on it and display us with results that is risk score and congestion score and no of vehicles that each class contains

5. Results

This chapter contains the outcome of each stage of the pipeline the aim of this chapter is to state what was measured

5.1 Detector performance

Fine tuning on the Indian dataset produced a mean average precision of 0.892 across the five vehicle classes, with 0.722 under the stricter averaging over IoU thresholds from 0.50 to 0.95. Precision reached 0.821 and recall 0.817

Measure	Value	Note
mAP50, all classes	0.892	Five-class scheme
mAP50-95, all classes	0.722	Averaged over IoU 0.50 to 0.95
Precision	0.821	At the best epoch
Recall	0.817	At the best epoch
Training duration	About 0.63 hours	30 epochs, single RTX 4060 laptop GPU
Retained training annotations	24,450	Explicit label mapping, no category dropped

Table 5.1 Detection performance of the fine-tuned model

Class	Train	Validation	Test	Total	Share
Auto-rickshaw	2,413	610	754	3,777	15.4%
Motorcycle	6,168	1,478	1,994	9,640	39.4%
Car	5,114	1,278	1,559	7,951	32.5%
Bus	997	258	275	1,530	6.3%
Truck	1,010	233	309	1,552	6.3%
Total				24,450	

Table 5.2 Annotation counts by class after label conversion

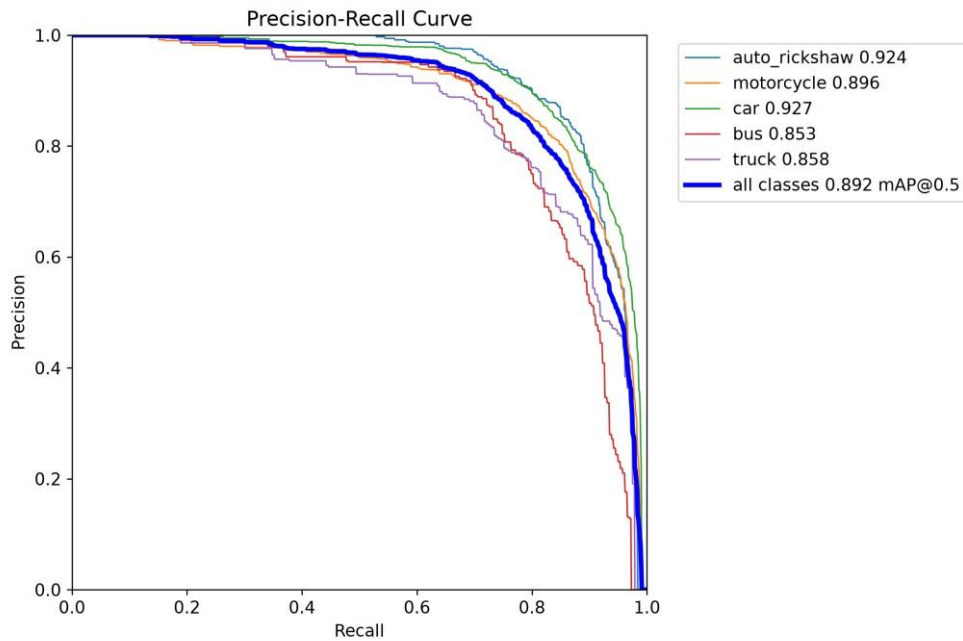


Figure 5.1 Precision-recall curve for each vehicle class on the validation split

5.2 Detection and tracking across the corpus

Applying the detector and tracking across all 124 videos of the training produced 4,477 Distinct tracked vehicles. The class distribution reflects the mix of Indian traffic rather than a general-purpose dataset two wheelers are the largest category and autorickshaw and cars also form substantial share

Class	Vehicles tracked	Share
Auto-rickshaw	646	14.4%
Motorcycle	1,721	38.4%
Car	1,479	33.0%
Bus	236	5.3%
Truck	395	8.8%
Total	4,477	

Table 5.3 Vehicles tracked across the corpus, by class

5.3 Congestion estimation

Windowing the videos at three seconds it produced 503 analysis windows drawn from the 124 videos. Cluster counts from two to five were fitted to the distribution of congestion values that is scored by silhouette.

Number of clusters	2	3	4	5
Silhouette score	0.825	0.800	0.579	0.542

Table 5.4 Silhouette score by cluster count for the congestion partition

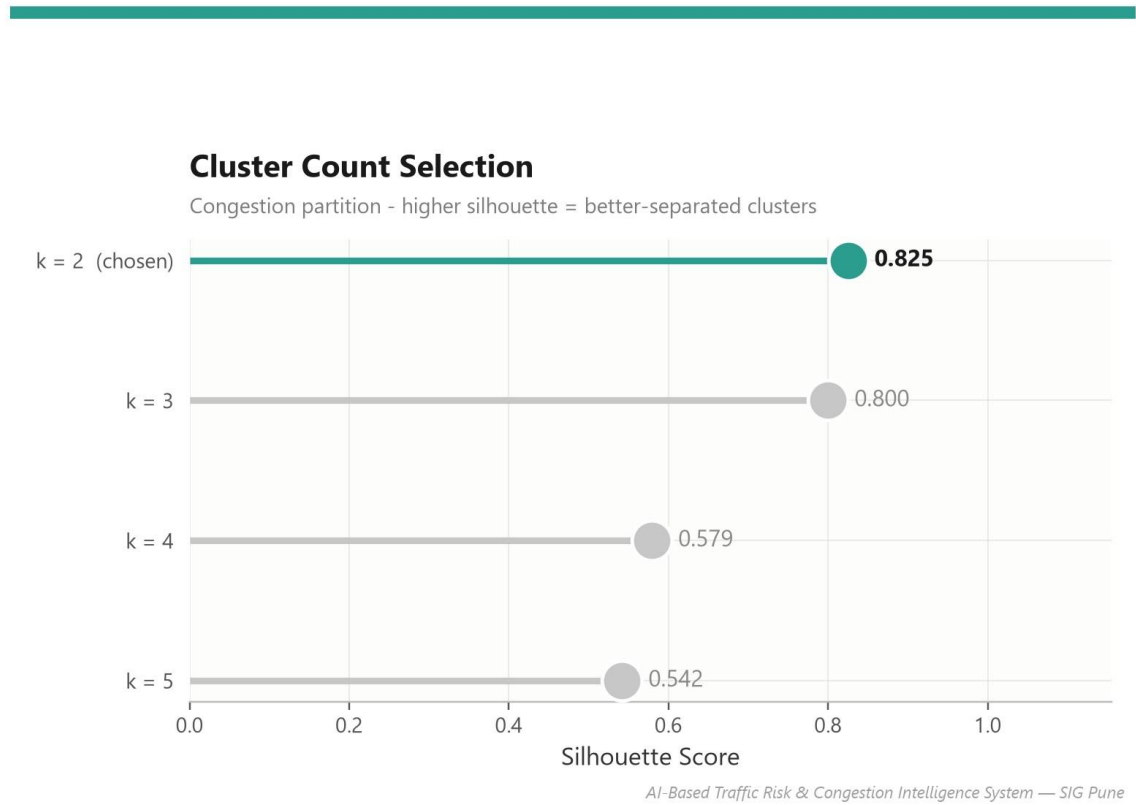


Figure 5.2 Silhouette score by cluster count for the congestion partition

Two clusters scored had the highest silhouette score of 0.825 so for congestion estimation two cluster were selected. The two clusters' centers fell at 6.06 and 25.43 and the midpoint was 15.75 this became her threshold boundary between low and high congestion

5.4 Risk descriptors and derived thresholds

Feature extraction produced 8,538 vehicle-window observations, each carrying the five descriptors defined in Section 4.6. Two thresholds were derived at this stage. Table 5.5 records both, together with the method used and the sample each rest on, since the derivation matters as much as the value.

Threshold	Value	Method	Evidence
Congestion boundary	15.75	Midpoint of two k-means cluster centres (6.06 and 25.43)	503 windows; silhouette 0.825 across k = 2 to 5
Proximity distance	0.2253 frame-widths	Midpoint of two k-means cluster centres (0.0844 and 0.3662 frame-widths)	324,909 nearest-neighbour distances; silhouette 0.758
Deceleration magnitude	0.2063 framewidths/s	Mean plus 1.5 standard deviations ($0.0421 + 1.5 \times 0.1095$)	156,804 observed frame-toframe speed drops

Table 5.5 Thresholds derived from the data and how each was obtained

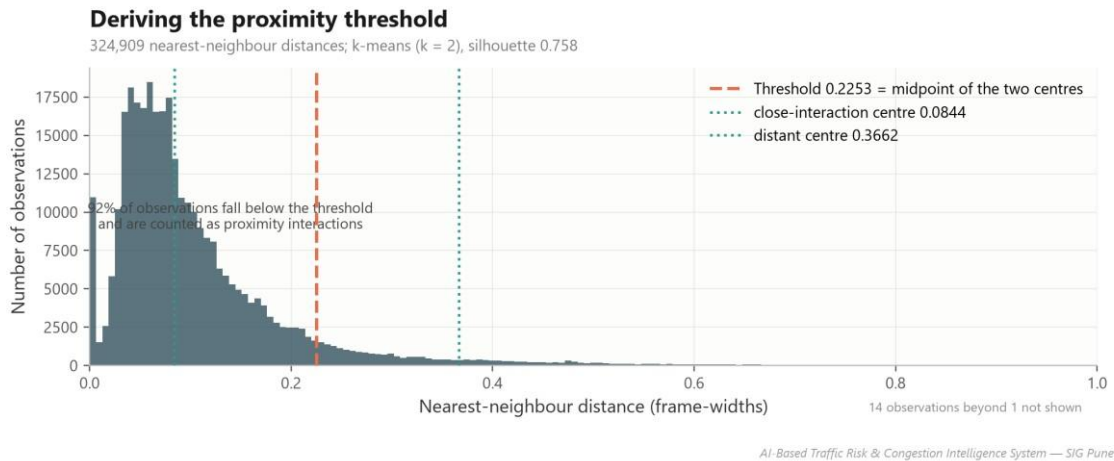


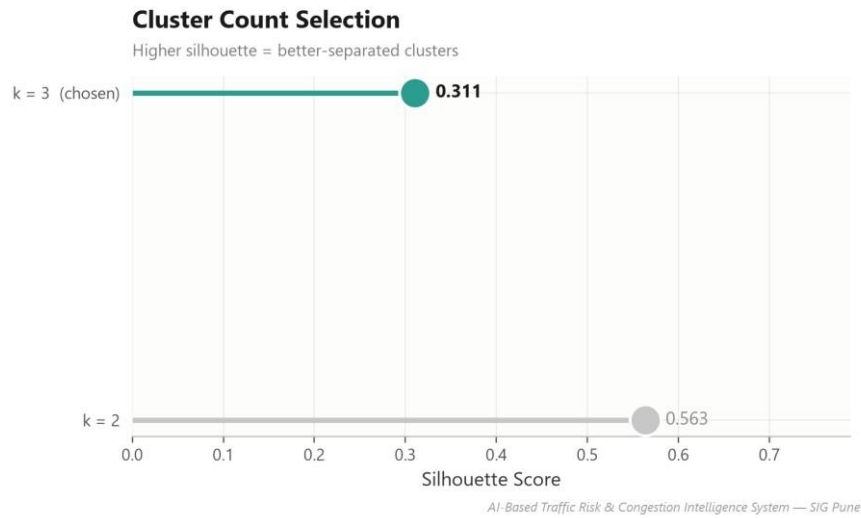
Figure 5.3 Distribution of nearest-neighbour distances and the derived proximity threshold

5.5 Risk partition

For risk partition k means was applied to the standardised descriptor matrix and using the silhouette score cluster count was selected. clusters of two and three were evaluated. The three cluster scores 0.311 whereas cluster two scored 0.563, so on silhouette alone two clusters separate the data better; three clusters were selected instead for the reason set out below, and were ordered by their mean and were labelled as safe, moderate and high risk

Risk class	Observations	Share
Safe	5,215	61.1%
Moderate	2,975	34.8%
High Risk	348	4.1%
Total	8,538	

Table 5.6 Distribution of vehicle-windows across the risk classes



Figure

5.4 Silhouette score for the candidate risk cluster counts

Silhouette score of 0.311 is a real but modest separation and is much weaker as compared to the silhouette score of 0.825 obtained for congestion. The congestion value is a single highest observed quantity with a binomial distribution as compared to the five features construct in which underlying regimes overlap. The margin between the two candidate partitions is wide at 0.253, but the two-cluster solution divides the data 95 per cent against 5 per cent: it isolates a small outlier tail and collapses everything else into a single safe bucket, leaving no middle band. Three clusters give a 61, 35 and 4 per cent split and the ordinal safe, moderate and high-risk scale that the rest of the pipeline depends on, so three were retained deliberately and both silhouette scores are reported so that the trade-off stays visible

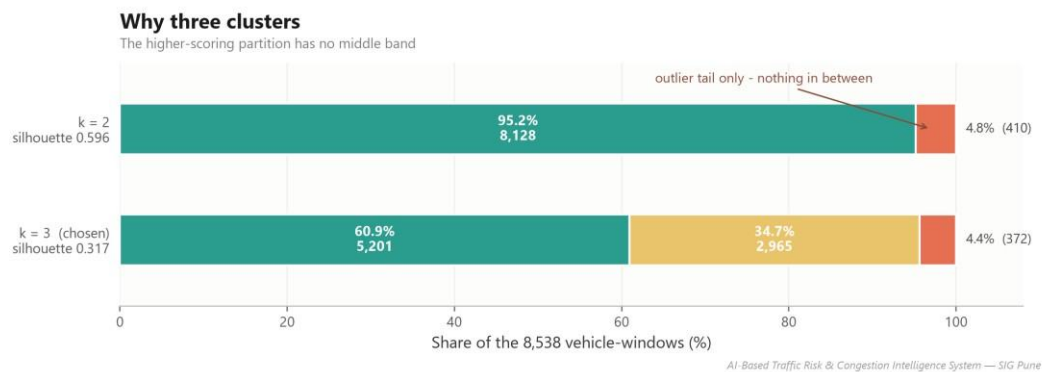


Figure 5.5 Cluster composition for the two candidate partitions

The 8,538 observations are divided across the three labels as shown in Figure 5.6: 5,215 Safe (61.1 per cent), 2,975 Moderate Risk (34.8 per cent) and 348 High Risk (4.1 per cent).

High Risk is a small minority class, which matters for the per-class results in the next section.

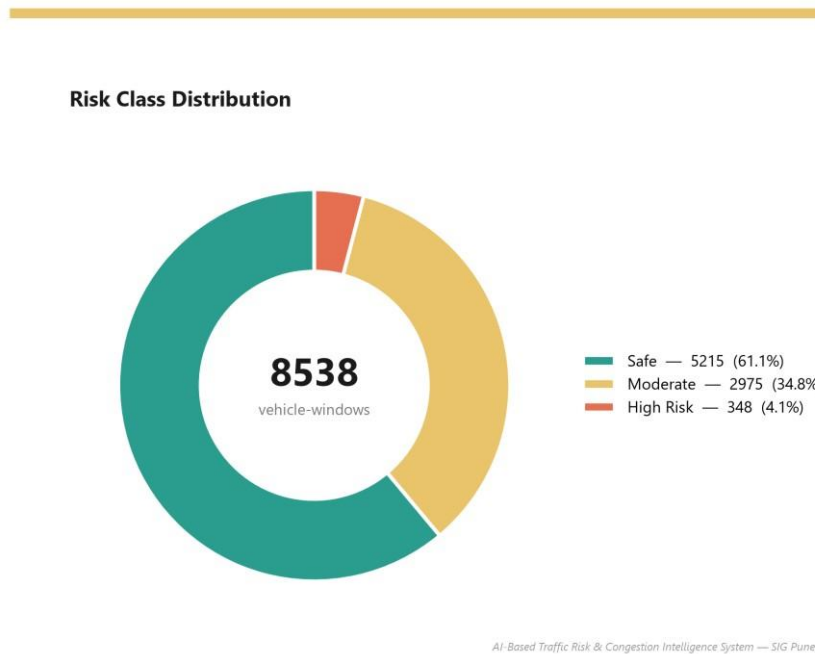


Figure 5.6 Distribution of the 8,538 vehicle-windows across the three risk classes

5.6 Supervised classification performance

For the train test split part 6,020 observations were in the training partition 983 observations in the validation and 1,535 observations in test. Both the classifiers were trained to predict the k means labels and automate the pipeline. The random forest reached a validation accuracy of 0.973 with a standard deviation of 0.010.

XG boost reached an accuracy higher than random forest at 0.991, misclassifying 14 of the 1,535 test observations against 33 for the Random Forest

Class	Support	Precision	Recall	F1
Safe	1023	0.987	0.983	0.985
Moderate	478	0.959	0.973	0.966
High Risk	34	1.000	0.912	0.954
Macro average	1535	0.982	0.956	0.968
Overall accuracy	1535	—	—	0.979

Table 5.7 Random Forest: per-class performance on the test partition

Class	Support	Precision	Recall	F1
Safe	1023	0.996	0.990	0.993
Moderate	478	0.979	0.992	0.985
High Risk	34	1.000	1.000	1.000

Macro average	1535	0.992	0.994	0.993
Overall accuracy	1535	—	—	0.991

Table 5.8 XGBoost: per-class performance on the test partition

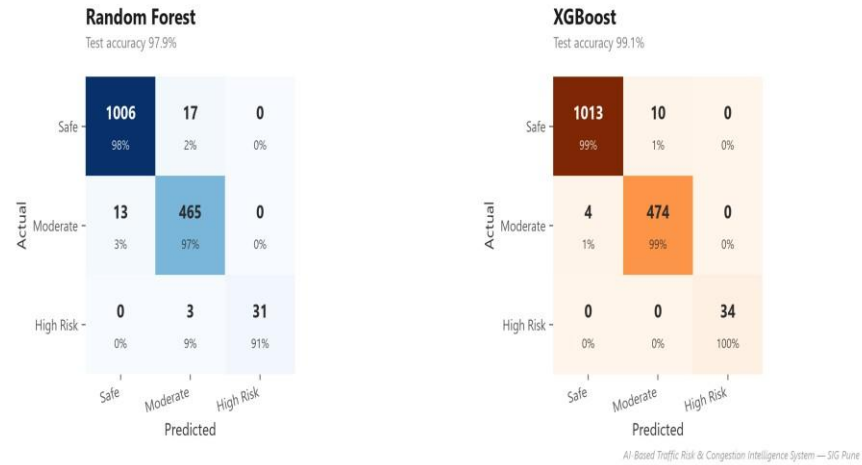


Figure 5.7 Test-set confusion matrices for Random Forest and XGBoost

By taking a look at the confusion matrix 1,006 out of 1,023 observations were classified as safe, 465 out of 478 were labelled as moderate risk and 31 out of 34 were classified as high risk

Cross-validation gave 0.973 against 0.979 on test, so the held-out videos were marginally easier than the training folds; with a fold standard deviation of 0.010 that difference is within ordinary variation. The gap between the weighted figure and the macro average is the part worth attention. Weighted by support the F1 is 0.979, but averaged equally across classes it falls to 0.968, and for High Risk alone to 0.954. That figure rests on 34 test observations, so it is estimated from a small sample and is correspondingly unstable: two further errors would drop it to roughly 0.92. The headline accuracy conceals this, which is one reason to read it with care.

5.7 Descriptor importance

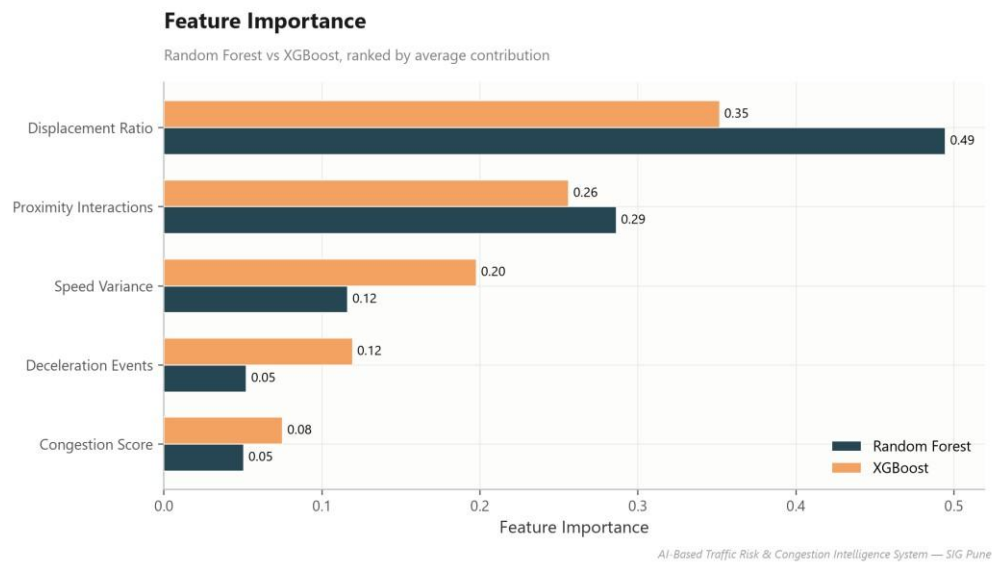


Figure 5.8 Feature importance reported by Random Forest and XGBoost

Both the classifiers Random Forest and XGboost give us a measure of how much each feature contributed. Both ranks displacement ratio first by a huge margin 0.495 for random forest and 0.352 for XGboost. Proximity interactions comes second in both with a score of 0.286 for the Random Forest and 0.256 for XG boost, followed by speed variance at 0.116 and 0.198 and deceleration events at 0.052 and 0.119. congestion score ranks last in both the methods at 0.051 and 0.075 respectively.

The main point to note is that ranking is dominated by movement descriptors rather than congestion. Congestion score contributes about five percent in the Random Forest and seven and a half percent in XGboost risk factor is not based on density. The two descriptors that matter, trajectory directness and proximity to other vehicles, are both behavioural, and both correspond directly to what Dong et al. (2023) and Boateng et al. (2023) identify as the standard markers of abnormal driving.

5.8 Per-clip results per-clip outputs were generated for videos spanning the three risk labels. Four are compared in Table 5.9, these are chosen because they cover the full frame rate and both the congestion states.

Clip	fps	Windows	Mean speed (frame-widths/s)	Risk profile
Traffic_in_Faridabad	25.0	24	0.296	Nineteen High Risk windows despite every window below the congestion boundary

Rickshaw_ride_..._Patna	25.0	96	0.226	High Risk clustered in bursts rather than spread evenly
12518849_1920_1080_60fps	59.94	10	0.114	Every window High Risk; congested throughout
13654799_1920_1080_30fps	30.0	7	0.047	No High Risk windows; congested in every window

Table 5.9 Four sample clips compared

The Faridabad clip and the 13654799 clip make the same point from opposite directions. Faridabad never exceeds four vehicles per window, comfortably below the congestion boundary, yet produces nineteen High Risk windows, while 13654799 sits above the boundary in every one of its seven windows yet produces none. If congestion were driving the risk label these results would be reversed. They are the clearest single piece of evidence that the descriptors capture something beyond crowding, and they agree with the importance ranking

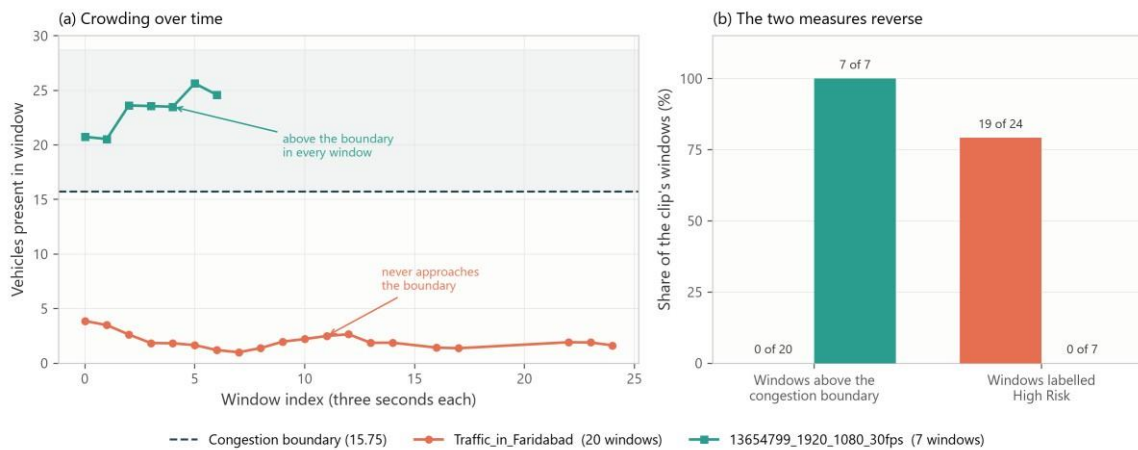


Figure 5.9 Crowding and risk compared for two contrasting clips

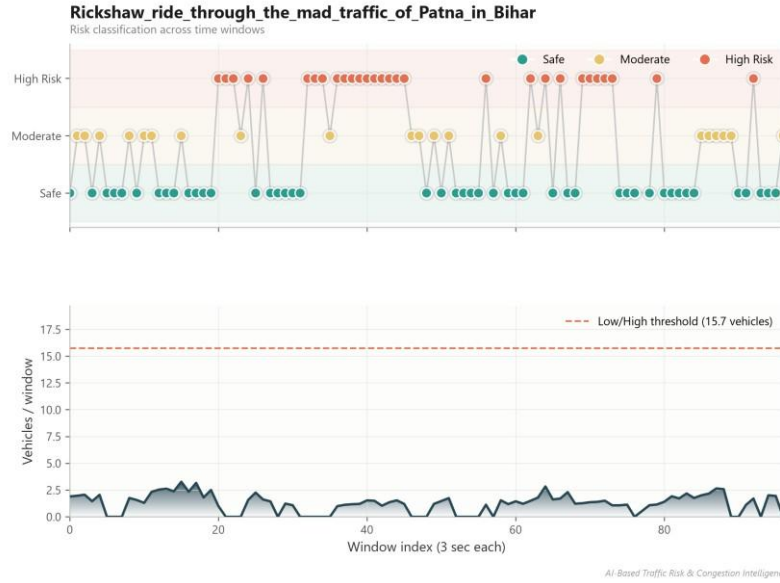


Figure 5.10 Risk classification and congestion timeline for a sample clip

The timeline shows the pattern that recurs across the corpus: High Risk windows arrive in short bursts rather than being spread evenly, which supports the argument of Thuremella et al. (2024) and Anis et al. (2024) that risk concentrates in brief episodes and is why the pipeline scores each window separately. Mean speeds differ substantially between clips, from 0.047 to 0.296 frame-widths per second across the four in Table 5.9. Because every displacement is divided by the frame width before a speed is formed, these figures are already independent of resolution, so the differences between clips reflect genuine differences in scene scale and camera framing rather than pixel counts. The figures remain relative rather than metric, a direct consequence of working without calibration, and the long tail of the speed distributions should be treated with caution because some very high apparent speeds are likely to be ego-motion rather than real vehicle movement.

5.9 Interactive dashboard

The Dash board was completed and it reads directly from the files, so every number it displays can be traced back to its initial stage of analytics. there are total 6 pages in the dashboard command center, model lab, congestion, video explorer, data explorer and analyze video where a user can upload a video that system has never seen and it will run the whole pipeline from congestion scoring to risk scoring without further input. The page returns an annotated replay of the clip, a timeline marking the windows classified as elevated risk with a sample frame drawn from each, and a summary card that can be downloaded.

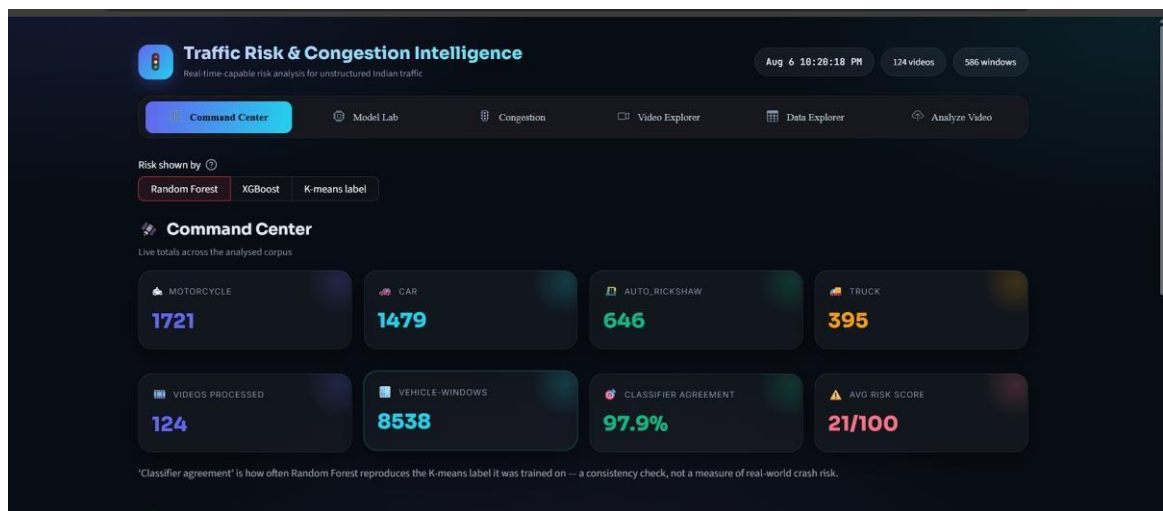


Figure 5.11 Dashboard overview

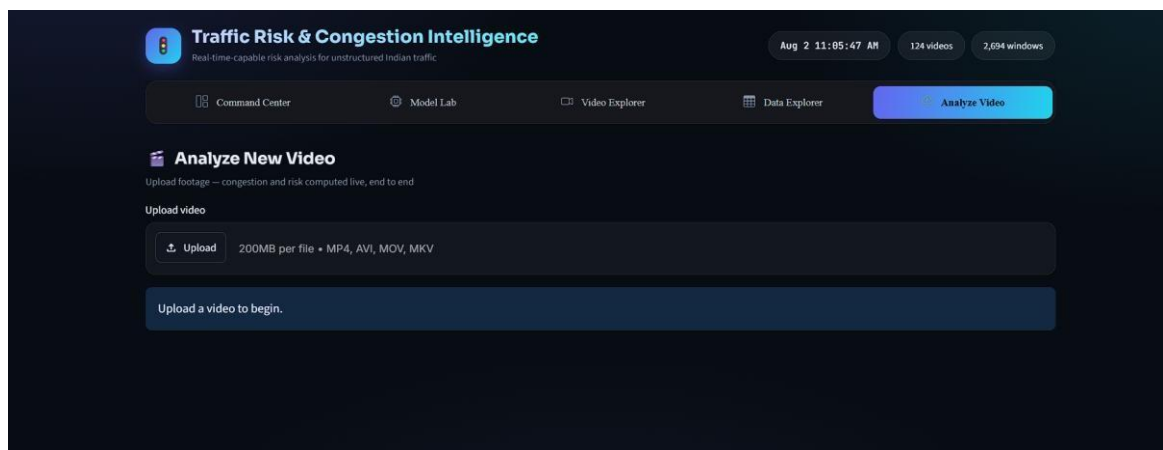


Figure 5.12 Analyze video page to upload a video

5.10 formulas Csv contents

Column	Role	Units
video_id	Source clip identifier	-
window_id	Index of the 3-second window	count
track_id	Tracker's identity for one vehicle	-
speed_variance	Speed steadiness	

deceleration_events	Count of sharp slowdowns	count
congestion_score	Vehicles sharing the window	count
cluster_idx	K-means cluster the record fell in	0, 1, 2
risk_label	Name given to that cluster	Safe / Moderate / High Risk
risk_composite_score	Mean standardised descriptor	standard deviations
rf_prediction	Random Forest's label	Safe / Moderate / High Risk
rf_confidence	Random Forest's probability	0-1
xgb_prediction	XGBoost's label	Safe / Moderate / High Risk
xgb_confidence	XGBoost's probability	0-1

Table 5.10 CSV file descriptor

5.10.1 Window Calculation:

Every record covers one vehicle over one window. A window is a fixed duration of real time, not a fixed number of frames.

$$N_window = \text{round}(T_window \times \text{fps})$$

Worked example

$$\begin{aligned} &12518849_1920_1080_60\text{fps} \\ &\text{fps} = 59.94 \\ &N_window = \text{round}(3.0 \times 59.94) = 180 \end{aligned}$$

$$\begin{aligned} &13275503_3840_2160_30\text{fps} \\ &\text{fps} = 30.00 \\ &N_window = \text{round}(3.0 \times 30.00) = 90 \end{aligned}$$

Reasoning –

The earlier version used a first 90 frames for every clip but it is true for videos at 30 fps but at 60 fps it's more than 30 fps clip so the formula was used where for each 3 second videos it was multiplied by its fps to decide the number of frames.

5.10.2 Displacement ratio:

Formula - $\text{displacement_ratio} = || p_{\text{last}} - p_{\text{first}} || / \text{Sum } || p_{(i+1)} - p_i ||$

Symbol	What it represents	Units	How it is obtained
p_i	Vehicle centre at its i-th appearance in the window	framewidths	Bounding-box centre (cx, cy) divided by the frame width
$p_{\text{first}}, p_{\text{last}}$	First and last position in the window	framewidths	Rows sorted by frame index
$ \cdot $	Euclidean distance	framewidths	$\text{sqrt}(dx^2 + dy^2)$
Numerator	Net displacement	framewidths	One distance, endpoints only
Denominator	Total path length	framewidths	Sum of all step lengths

Table 5.11 Displacement formula terms

If the value is 1 then it means that the vehicle is in a perfectly straight line since the two distances coincide and fall towards 0 as the path bends the quantity is ratio of two lengths so the units cancel each other.

```
12518849_1920_1080_60fps    window    0    track    17
displacement_ratio = 0.663415
the vehicle covered about 66% of its travelled distance as net progress
```

```
12518849_1920_1080_60fps    window    0    track    9
displacement_ratio = 0.016074
almost all motion cancelled out - the vehicle ended where it began
```

6. Discussion

The most important number in this report is the test accuracy which came to be about 0.979 it was achieved by the two models on the videos that neither of the classifiers had seen during training. To set out plainly what it does and what it does not establish is the labels that the classifiers were trained to predict were not generated by human judgement. They came from k-means clustering of five descriptors the classifiers then received as input, so the models were asked to recover a partition of a feature space from the coordinates of that space. That both succeeded tells us something real that the boundaries are geometrically regular, they hold across different videos, and a tree ensemble can represent them. It does not tell us that the partition corresponds to what a traffic engineer would call risk. Had the clustering split the data along an axis with no safety meaning at all, an equally high accuracy

would still have been obtainable. The figure measures self-consistency, not validity, and I would not defend it as anything more.

What reasons support the claim that partition is meaningful is that the five features that were chosen as the established safety indicators point to them and the ordering of the clusters by their mean descriptor profile runs in the direction those indicators predict. That evidence is reasonable but indirect. The evaluation protocol was built to make the accuracy figure as informative as it can be under these constraints the split was made at video level so the test partition contains no road the models had seen, the standardiser and the clustering model were fitted on the training partition alone so that no property of the held-out data influenced its own labels, and two independently regularised model families were used so the result is not an artefact of one algorithm. Within the limits of unsupervised labelling the protocol is sound. The limit is the labelling, not the protocol, and closing it would need expert annotation rather than a better classifier.

The strongest finding found in this project was that risk partition is not simply tracking traffic density. Three separate evidences point towards this. Congestion score ranks last in both importance measures, at 0.051 in the Random Forest and 0.075 in XGBoost, which is close to nothing. The Faridabad clip produces nineteen High Risk windows while never once crossing the congestion boundary. The 13654799 clip sits above that boundary in all seven of its windows and produces none. If crowding were driving the risk label, those two results would be the other way round. Taken together they rule out the most obvious failure mode, which was that the system had learned an expensive way to count vehicles, and they matter because that is exactly what separates this work from the density-estimation. The two clustering results are worth comparing directly. Congestion clustering reached a silhouette of 0.825 on a single directly observed variable, while risk clustering reached only 0.311 on a five-dimensional construct. The gap says that congestion is genuinely bimodal in this corpus, with a real boundary between free-flowing and crowded conditions, whereas risk varies continuously and the three-way division imposed on it is a convenience rather than a discovery. A continuous risk score with indicative bands would represent that evidence more honestly than a hard label does, and would behave better for observations sitting near a boundary. I would treat the three-class output as a presentational choice rather than a claim about the structure of the data.

BoT-SORT was chosen specifically for that component and it did not work on a single video, because the sparse optical flow it depends on assumed small inter-frame

displacement and geometric consistency, and footage shot from a two-wheeler violates both. This is not a defect in the tracker so much as a mismatch between the data regime a method was validated on and the regime it was applied to. The consequence is bounded but real: without compensation the motion model attributes the camera’s own movement to the tracked vehicles, which inflates apparent displacement and pushes shake into the speed signal. Displacement ratio and speed variance are both computed from that signal, and since displacement ratio is the dominant descriptor, this is not a small caveat. The descriptors remain comparable within a clip, because every vehicle in a clip shares the same camera motion, but comparison across clips is weakened.

The second defect is a window-length inconsistency between two stages, which I found while checking the congestion output against the per-clip figures. The congestion stage defines its window in seconds and converts to frames using each video’s measured rate, giving 75 frames at 25 fps, 90 at 30 fps and 180 at 59.94 fps, all exactly three seconds. The feature stage instead uses a fixed 90-frame window. For the 116 videos recorded at 30 fps the two coincide exactly, but for the remaining eight they diverge, producing windows of 3.6 seconds at 25 fps and 1.5 seconds at 59.94 fps. The effect on the aggregate results is small, since those videos are about six per cent of the corpus, but it is a genuine reproducibility flaw and the fix is to define the feature window in seconds as the congestion stage already does.

the detection results sit within the range reported for fine-tuned single-stage detectors on Indian imagery, and the auto-rickshaw performance supports the argument that Indian road scenes need domain-specific training data rather than transfer from Western datasets. On congestion the departure is deliberate: the density-based methods reviewed in Chapter 2 almost always fix a count threshold whose value is rarely justified beyond the authors’ judgement, and deriving it by clustering removes that arbitrariness and make the procedure, rather than the number, the thing that transfers. On risk the departure is larger. Yang and Hong (2026) show what is achievable when calibration is available, reaching over ninetyone per cent precision with a ground-plane projection. This project gives up metric space entirely in exchange for working on uncalibrated video. What is gained is that the method runs on any road footage without survey work; what is lost is comparability of absolute values across camera configurations, visible in Chapter 5 where speeds can be given only in frame-widths per second, and the ability to state a result in physically interpretable units. Whether that trade is worth making depends on the application. For

screening large volumes of uncalibrated footage, it clearly is, and for a detailed study of a single junction it clearly is not.

The limitations of the study are summarised in Table 6.1. Two of them deserve emphasis. The absence of pedestrians is the most consequential restriction of scope, because pedestrian conflicts account for a large share of serious road injuries in Indian urban environments and the system cannot see them at all. And the minority status of the High-Risk class means that the category which matters most operationally is the one the system predicts least reliably, resting as it does on thirty-four test cases.

Limitation	Effect	Handling
Risk labels from clustering, not annotation	Accuracy measures self-consistency, not validity	Stated throughout; expert annotation proposed
Camera motion compensation failed on all videos	Ego-motion contaminates the two dominant descriptors	Reported openly; pre-stabilisation or dense flow proposed
Window length in frames, not seconds, at the feature stage	Affects the 8 videos not recorded at 30 fps	Documented in Chapter 6; one-line fix identified
No scene calibration	Pixel speeds not comparable across resolutions	Accepted deliberately; thresholds re-derivable per corpus
High Risk is 4.1 per cent of observations	F1 of 0.954 rests on 34 test cases	Macro average reported alongside accuracy
Modest risk-cluster silhouette (0.311 against 0.563)	Three-way division imposed rather than natural	Reported plainly; continuous scoring proposed
Pedestrians and animals not modelled	A significant class of conflicts is invisible	Scope stated in Chapter 1; extension proposed

Table 6.1 Limitations and how each was handled

Despite these limitations the system supports uses that traditional crash-based analysis cannot.

7. Conclusion

7.1 Summary

This project set out to determine whether ordinary, uncalibrated video of unstructured Indian traffic could be converted automatically into interpretable congestion and risk assessments, using thresholds derived from the data and hardware available to one student. A seven-stage pipeline was designed, built and evaluated to answer that question.

A published fine-grained annotation set was converted to a five-class coarse scheme and used to fine-tune a lightweight detector, which reached 0.892 mAP50 with the autorickshaw

class at 0.924. The detector was paired with a tracker and applied across 124 videos. Track records were divided into three-second windows, and clustering selected a two-cluster congestion partition with a silhouette of 0.825, giving a boundary of 15.75 that labelled 458 of 503 windows Low and 45 High. Five calibration-free descriptors were computed for 8,538 vehicle-window observations, the proximity cut-off derived by clustering 324,909 nearest-neighbour distances and the deceleration cut-off set at 1.5 standard deviations above the mean of 156,804 speed drops. Three-way clustering on the training partition alone generated ordinal risk labels, which two classifiers reproduced on 1,535 held-out observations with a test accuracy of 0.979 and a macro F1 of 0.968.

7.2 Objectives revisited

All six objectives of the project were met: the dataset was converted and verified against the published annotation count, the detector fine-tuned and quantified, detection and tracking integrated across the corpus, congestion estimated from a clustered threshold, five descriptors engineered and validated by two classifiers under a leakage-controlled protocol, and the dashboard built on real pipeline artefacts. Returning to the principal research question, the answer is a qualified yes. A pipeline of this kind can produce interpretable congestion and risk assessments from uncalibrated video without manual risk annotation. The congestion component is on firm ground, because its threshold is recovered from a genuinely bimodal distribution with a strong silhouette. The risk component is on softer ground: it produces a stable, learnable partition driven by behavioural rather than density features, which is encouraging and is supported by the two contrasting clips in Section 5.8, but whether that partition matches risk as a traffic engineer would assess it remains unestablished on the evidence gathered here.

7.3 Future Scope

Four extensions follow from the limitations in Table 6.1, in descending order of importance.

1. Expert annotation of a validation subset. A sample of several hundred windows, stratified across the three predicted labels and reviewed independently by two or more experienced assessors, would allow agreement between the pipeline's output and human judgement to be computed. This one addition would convert every accuracy figure in this report from a statement about internal consistency into a statement about validity, and it is by a wide margin the highest-value extension available.

2. Resolving the camera motion problem. Either pre-stabilising the footage or replacing the sparse optical flow estimator with a dense one should restore the egomotion correction that failed here and remove a known contaminant from the two dominant descriptors.
3. Extending detection to vulnerable road users. Adding pedestrian and animal categories, which needs a training set carrying those annotations, would bring into scope a class of conflict the present system cannot observe and which accounts for a large share of serious injury on Indian roads.
4. Replacing hard labels with a continuous score, and defining the feature window in seconds rather than frames. The first reflects the modest cluster separation more honestly; the second removes the inconsistency documented in Chapter 6.

8. References

References follow the style of the American Psychological Association.

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Annexure

Annexure A: Software artefacts and data products

The following scripts make up the implementation. Each is independently executable and consumes only files written by earlier stages.

Script	Stage	Function and output
convert_fgvd.py	1	Collapses the label hierarchy; writes YOLO-format annotations and the split

finetune.py	2	Fine-tunes YOLOv11m; writes weights and training summaries
detect_and_track.py	3	Applies the detector with BoT-SORT; writes one track-record CSV per video
build_fps_lookup.py	-	Measures every source frame rate; writes video_fps_lookup.json
congestion_analysis.py	4	Segments into three-second windows; writes congestion_windows_labeled.csv, congestion_video_summary.csv, congestion_thresholds.json
features_risk.py	5	Computes the descriptors; writes risk_features.csv and risk_feature_thresholds.json
risk_model.py	6	Pseudo-labels, video-level split, training and evaluation; writes risk_predictions.csv and risk_model_metrics.json
generate_charts.py	7	Produces all figures and the caption index
dashboard.py, theme.py	7	Interactive Streamlit console and shared visual definitions

Table A.1 Software artefacts and data products

Annexure B: Derived parameters

Parameter	Value	Source
Congestion clusters	2 of 2–5 evaluated	Silhouette 0.825, 0.800, 0.579, 0.542
Congestion centres	6.06 and 25.43	k-means, random seed 42
Congestion boundary	15.75	Midpoint of the two centres
Proximity centres	0.0844 and 0.3662 frame-widths	k-means on 324,909 nearest-neighbour distances
Proximity threshold	0.2253 frame-widths	Midpoint; silhouette 0.758
Deceleration threshold	0.2063 frame-widths/s	Mean 0.0421 + 1.5 × SD 0.1095, from 156,804 samples
Risk clusters	3 of 2–3 evaluated	Silhouette 0.311 against 0.563
Partition sizes	1,726 / 453 / 515	60/20/20 applied at video level

Table B.1 Derived parameters

Annexure C: Reproducibility measures

Every derived threshold is written to a parameters file rather than embedded in code, and random seeds are fixed for all stochastic operations, so the values used in any run can be recovered from its outputs. All model metrics are serialised to risk_model_metrics.json, from which both the report tables and the dashboard read, so the two cannot disagree.

Annexure D: Frontend

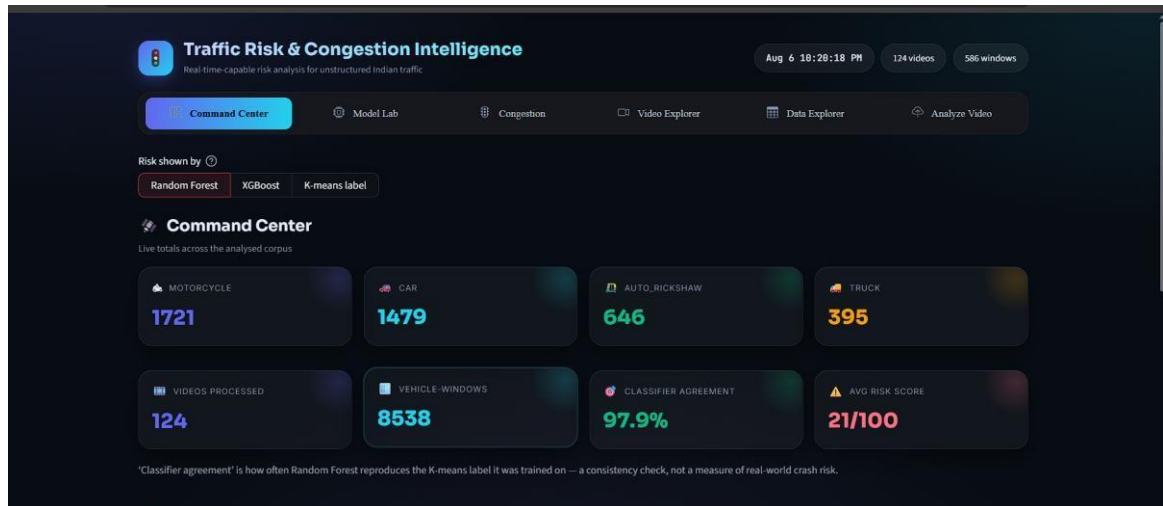


Figure D.1 Command Centre page showing corpus-wide totals and classifier agreement

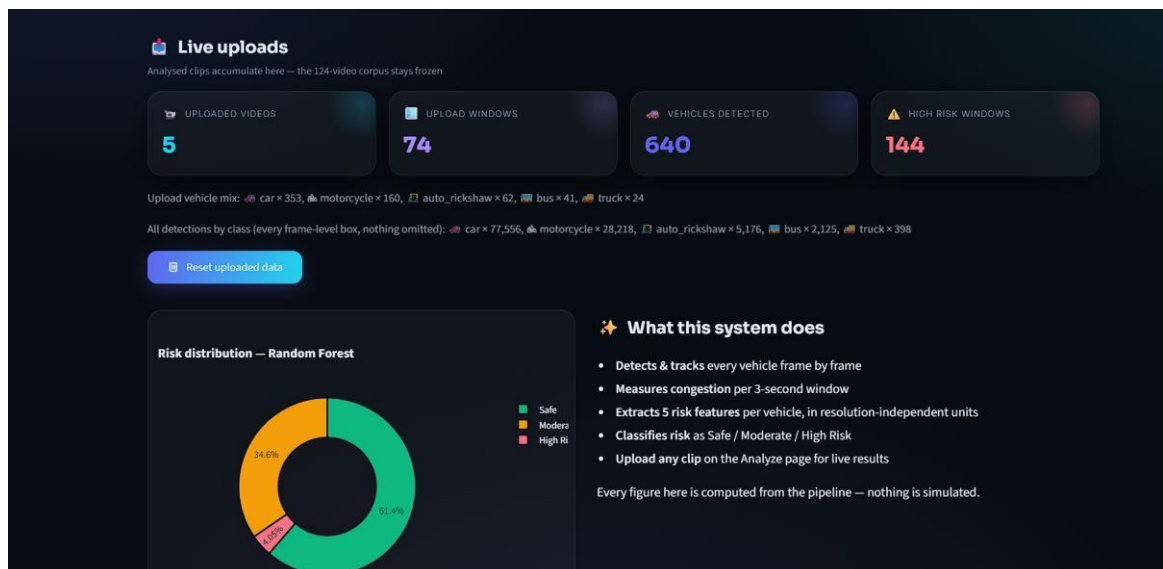


Figure D.2 Live uploads panel with the risk distribution for analysed clips

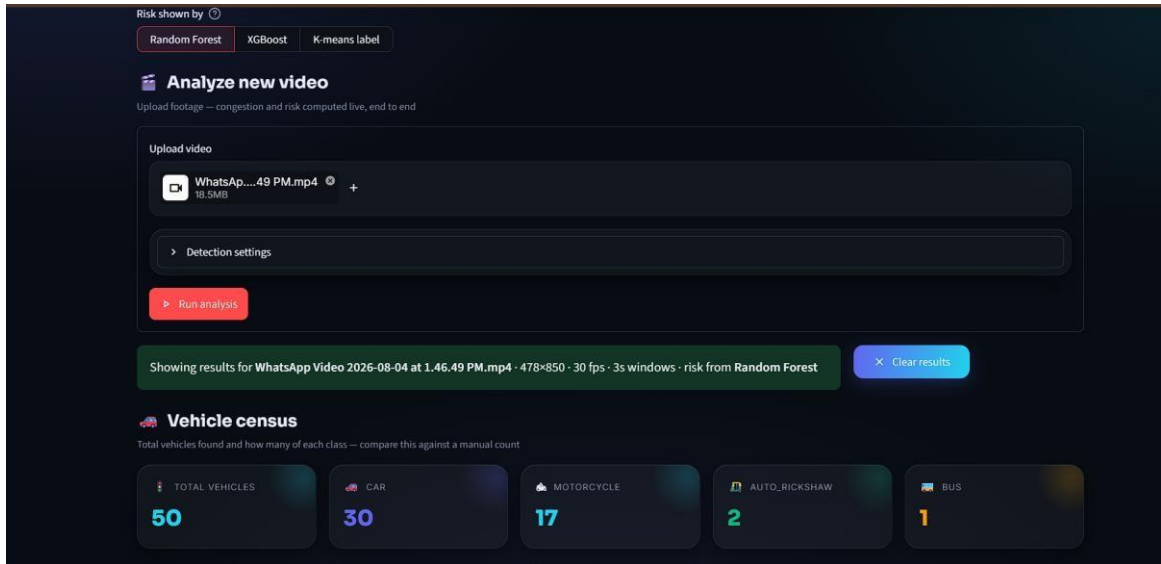


Figure D.3 Analyze video page with the vehicle census for an uploaded clip

Appendix

Appendix A: Data schemas

The track-record files written by Stage 3 carry one row per tracked vehicle per frame, with the columns frame, track_id, class_id, class_name, cx, cy, width, height and confidence, where cx and cy are the bounding-box centre coordinates in pixels. The descriptor table written by Stage 5 is the central data product of the pipeline and has the following columns. Its columns are video_id and window_id identifying the source clip and the window within it, track_id identifying the vehicle, and the five descriptors displacement_ratio, speed_variance, proximity_interactions, deceleration_events and congestion_score. The assigned class of Safe, Moderate or High Risk is held in the risk_label column of the Stage 6 prediction file rather than in this table. Proximity interactions count frames spent within 0.2253 frame-widths of another vehicle and deceleration events count speed drops exceeding 0.2063 frame-widths per second.

Appendix B: Order of execution

Reproducing the results requires the stages to run in order, each completing before the next begins: convert_fgvd.py to prepare the detection dataset and check the retained annotation count against the published figure; finetune.py to fine-tune the detector, whose weights feed

the next stage; `detect_and_track.py` to process the corpus, which is safe to interrupt and resume; `build_fps_lookup.py` to measure every source frame rate; `congestion_analysis.py` to derive the congestion boundary; `features_risk.py` to compute the descriptors and derive the proximity and deceleration thresholds; `risk_model.py` to generate pseudo-labels, split at video level and train and evaluate; and `generate_charts.py` to produce all figures and the caption index. The dashboard is launched separately with `dashboard.py`, which imports `theme.py` automatically; `theme.py` is never run directly.

Proforma 4

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1. Introduction India consists of very diverse traffic flows. Vehicles of different classes like cars, autorickshaws, buses, bicycles etc are on the same road stretch with usually little or almost no lane discipline. The movement of vehicles is continuous and cannot be divided or organised into lanes. Such kind of movements that usually do not follow lane discipline is unstructured traffic and traditional methods do not work where they assume several standards are followed. Lane occupancy cannot be calculated using standard methods where the lane discipline is not consistently followed, we in this study have tried to create and identify a solution that helps us build analytical models by combining different computer vision, tracking and machine learning models.

2. Literature Review In 2022 Zand and team used ByteTrack and found that even weak detection that are removed can still hold enough information of a vehicle. Aharon and team in the same year published BoT-SORT it is a multi object tracking algorithm it combines camera motion compensation, appearance information and utilises a Kalman filter. The selection of detectors is fine but for it to perform optimally training it on right conditions is important. In 2019 Varma and team used the India Driving Dataset and found that European driving models don't perform well in Indian roads. Varma and team achieved an accuracy of .09 on Indian vehicles. The FGVD dataset that was published by Khoba and team in 2022 arranges vehicles in different classes making identification of vehicles on Indian road condition easier. Traditional threshold techniques and numbers cannot be used for Indian roads cannot be used directly as traditional methods assume certain rules are followed so we need a method.

In 1967 MacQueen introduced the k means clustering that assigns each one of the observations to its cluster that is closest to its mean. Since Kmeans itself cannot choose the number of clusters in 1987 Rousseeuw came up with Silhouette coefficient which weighs an observation's closeness to its own cluster against the nearest alternatives. Onelcin and Alver built a weaving-section score for occasional weaving at fixed merge points, which shows by contrast that Indian traffic needs weaving treated as a constant condition rather than an event. Khan and their team observed in 2024 that fatigue has drawn far more research attention than sudden lateral movement, naming precisely the gap this